

A power-compact weighted-shift subcase of the positive-commutator radical problem

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Run fa_banach_001, lane 0

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Abstract

Kandić and Šivic asked whether their theorem for two positive power-compact operators remains true when only one factor is power compact: must some power of a positive commutator belong to the radical of the Banach algebra generated by the two factors? We prove an affirmative subcase. If the power-compact factor is a unilateral forward weighted shift with strictly positive weights on c_0 or ℓ^p , then the $3m$ -th power of the commutator is radical whenever the m -th power of the shift is compact. The key new observation is that order semicommutation forces the other positive operator to preserve the shift's maximal tail nest.

1 The source question

Let E be a Banach lattice, let $A, B \in \mathcal{L}(E)$ be positive, and put $C = AB - BA \geq 0$. The source paper proves that if both A and B are power compact, then some power of C belongs to the radical of the Banach algebra generated by A and B [3, Theorem 4.7]. It ends by asking whether only one of A, B need be power compact.

Remark 5.6. *Suppose that A and B be positive operators on a Banach lattice E with a positive commutator $AB - BA$. If one of the operators is power compact, then $AB - BA$ is quasi-nilpotent by [5 Corollary 3.2]. In particular, this implies that a positive commutator $AB - BA$ of positive power compact operators A and B is quasi-nilpotent which also follows from Theorem [4.7](#).*

We finish this paper with two open questions.

- (a) How big is the Banach algebra generated by $\langle V \rangle \cup [V]$?
- (b) Can we assume in Theorem [4.7](#) that only one of the operators A and B is power compact?

The earlier theorem of Drnovšek already shows that if only one factor is power compact, then C is power compact, ideal-triangularizable, and quasinilpotent [2, Corollary 3.2]. What is missing in general is control of products DC^n for arbitrary D in the algebra generated by A and B .

2 Statement of the partial result

Let E be either $c_0(\mathbb{N}_0)$ or $\ell^p(\mathbb{N}_0)$, $1 \leq p < \infty$, over $\mathbb{R}$ or $\mathbb{C}$. Write $(e_j)_{j \geq 0}$ for its standard unit vectors.

Theorem 1. *Let A be the positive unilateral forward weighted shift*

$$Ae_j = \alpha_j e_{j+1} \quad (j \geq 0), \quad \alpha_j > 0.$$

Assume that A^m is compact for some $m \geq 1$. If $B \geq 0$ and $AB \geq BA$, set $C = AB - BA$ and let $\mathcal{A} = \text{Alg}(A, B)$ be the norm-closed Banach algebra generated by A and B . Then

$$C^{3m} \in \text{Rad}(\mathcal{A}).$$

The exponent is explicit but is not claimed to be optimal.

3 The forced tail nest

For $n \geq 0$ define the closed tail ideal

$$M_n = \overline{\text{span}}\{e_j : j \geq n\}.$$

Together with $\{0\}$, these spaces form a complete maximal chain: successive quotients M_n/M_{n+1} are one dimensional and $\bigcap_n M_n = \{0\}$.

Lemma 2 (boundary propagation). *Under the hypotheses of Theorem 1, every M_n is invariant under B . Consequently A , B , and every member of $\mathcal{A}$ are triangular with respect to the tail chain. The diagonal coefficients of C on this chain are all zero.*

Proof. Write

$$b_{ij} = e_i^*(Be_j) \geq 0.$$

The (i, j) matrix coefficient of $AB \geq BA$ says

$$0 \geq \alpha_j b_{0,j+1} \quad (i = 0), \quad (1)$$

$$\alpha_{i-1} b_{i-1,j} \geq \alpha_j b_{i,j+1} \quad (i \geq 1). \quad (2)$$

All weights are strictly positive. Equation (1) therefore gives $b_{0k} = 0$ for every $k > 0$. Suppose inductively that $b_{i-1,k} = 0$ whenever $k > i - 1$. If $k > i$, apply (2) with $j = k - 1$; its left side is zero, so $b_{ik} = 0$. Thus

$$b_{ik} = 0 \quad (k > i).$$

Equivalently, $B(M_n) \subseteq M_n$ for every n . The same is clear for A , and tail invariance is preserved by algebraic operations and norm limits.

Finally, for $n \geq 1$,

$$(AB)_{nn} = \alpha_{n-1} b_{n-1,n} = 0, \quad (BA)_{nn} = \alpha_n b_{n,n+1} = 0;$$

the $n = 0$ case follows from (1). Hence every diagonal coefficient of C is zero. $\square$

4 A compact power and the radical criterion

We use the standard Aliprantis–Burkinshaw domination theorem: if $0 \leq S \leq T$ on a Banach lattice and T is compact, then S^3 is compact [1, Theorem 5.14]. This is the same domination result used in the proof of [2, Corollary 3.2].

Lemma 3. *Under the hypotheses of Theorem 1, C^{3m} is compact.*

Proof. Because $0 \leq C \leq AB$, positivity and induction give $0 \leq C^m \leq (AB)^m$. The inequality $BA \leq AB$ similarly permits each B to be moved to the right, giving

$$(AB)^m \leq A^m B^m.$$

The positive operator $A^m B^m$ is compact. Thus $0 \leq C^m \leq A^m B^m$, and the domination theorem yields $(C^m)^3 = C^{3m}$ compact. $\square$

Proof of Theorem 1. Pass to the standard complexification in the real case. Let $D \in \mathcal{A}^\#$, the unitization of $\mathcal{A}$. Lemma 2 shows that D and C are triangular on the maximal tail chain, and the diagonal of C is zero. Therefore every diagonal coefficient of DC^{3m} is zero. By Lemma 3, this operator is compact. Ringrose’s theorem for compact operators and complete triangularizing chains now gives

$$\sigma(DC^{3m}) = \{0\}.$$

Thus $r(DC^{3m}) = 0$ for every $D \in \mathcal{A}^\#$. The spectral-radius characterization of the Jacobson radical proves $C^{3m} \in \text{Rad}(\mathcal{A})$. $\square$

5 Scope and obstruction to the full result

The proof separates the source problem into two issues. Power compactness of the commutator is not the main obstruction: from one compact power of A , order domination already supplies the compact power C^{3m} . The difficult point is to make every product DC^{3m} quasinilpotent. For a forward weighted shift, the boundary row in (1) propagates through the matrix and creates a common maximal nest. A general positive power-compact operator has no analogous coordinate boundary, so the argument does not settle the full question.

If the power-compact factor is actually nilpotent, say $A^m = 0$, the same inequalities give $0 \leq C^m \leq A^m B^m = 0$, and the source conclusion is trivial. The theorem above is useful precisely when A is power compact but not nilpotent.

References

- [1] C. D. Aliprantis and O. Burkinshaw, *Positive Operators*, Springer, 2006.
- [2] R. Drnovšek, Once more on positive commutators, *Studia Math.* **211** (2012), 241–245; [arXiv:1211.0812](#).
- [3] M. Kandić and K. Šivic, On the positive commutator in the radical, *Positivity* **21** (2017), 99–111; [arXiv:1405.4988](#).